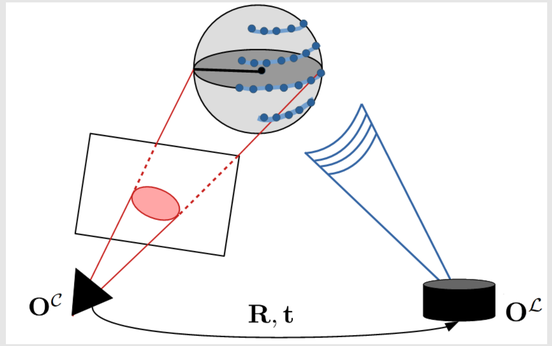
Automatic Extrinsic Calibration of LiDAR-Camera Setup Using a Spherical Target

Tekla Tóth, Levente Hajder

Eötvös Loránd University, Department of Algorithm and Applications

Schematic setup

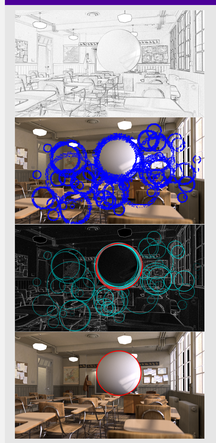
Calibration using spherical object



Intrinsic parameters
- Focal length
- Principal point (center of the image)
- Pixel scale
- Distorsion

Extrinsic parameters
- Location (translation)
- Orientation (rotation)

Ellipse detection from image



1. **Edge magnitudes** after applying the Sobel operator.
2. **Candidate points** for ellipse fitting. After edge-based circle fitting, the strongest edge points are selected radially for the candidate circles.
3. **Candidate ellipses.** Red color denotes the circle with the highest score.
4. **Final fitted ellipse.** The outputs of the fitting are the ellipse coefficients.

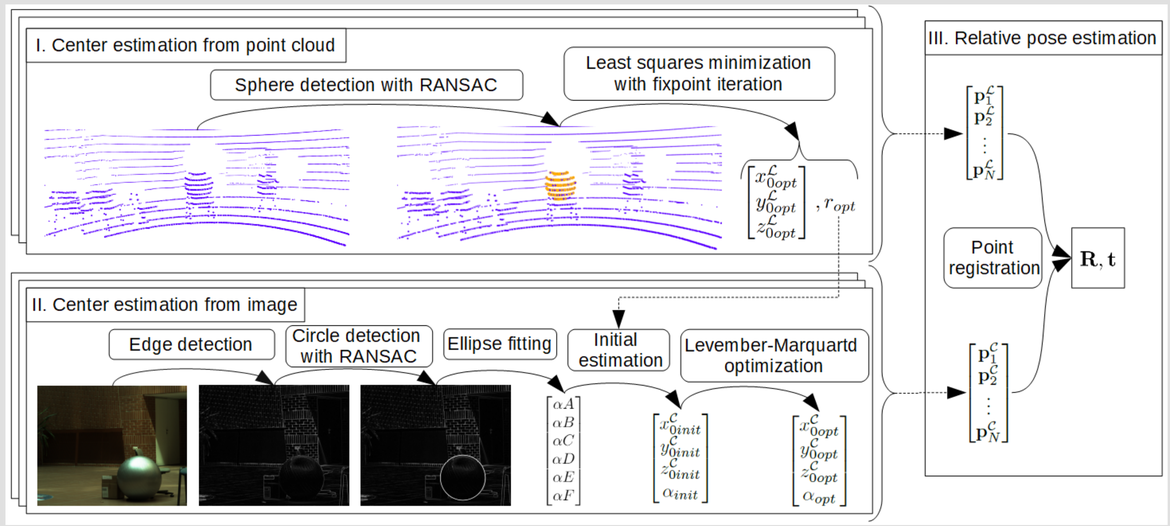
Initial and optimal sphere center estimation based on the implicit form of a conic section

$$A\hat{u}^2 + B\hat{u}\hat{v} + C\hat{v}^2 + D\hat{u} + E\hat{v} + F = 0,$$

where the coefficients are

$A = r^2 - y_0^2 - z_0^2$, $B = 2x_0y_0$, $C = r^2 - x_0^2 - z_0^2$,
$D = 2x_0z_0$, $E = 2y_0z_0$, $F = r^2 - x_0^2 - y_0^2$.

Calibration Pipeline



Mean error and standard deviation of optimal sphere center estimation

- LiDAR data – *Gaussian noise added with varying mean (μ) and standard deviation (σ)*

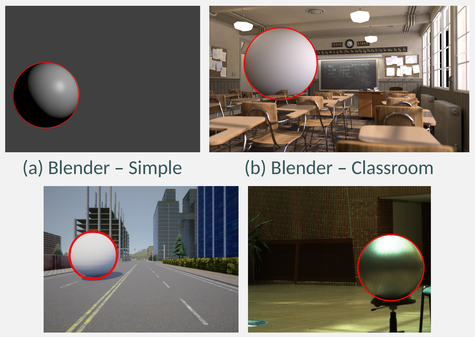
	In terms of $d^{\mathcal{L}}_{opt}$ with varying μ				
μ	0.00	0.02	0.04	0.06	0.08
avg. (m)	0.005	0.022	0.047	0.072	0.0961
std. (m)	0.003	0.004	0.006	0.006	0.004

	In terms of $d^{\mathcal{L}}_{opt}$ with varying σ					
σ	0.00	0.02	0.04	0.06	0.08	0.1
avg.	0.004	0.005	0.007	0.041	0.090	0.185
std.	0.003	0.003	0.003	0.011	0.011	0.020

- Camera images

	In terms of							
	$x^{\mathcal{C}}_{0init}$	$y^{\mathcal{C}}_{0init}$	$z^{\mathcal{C}}_{0init}$	$d^{\mathcal{C}}_{init}$	$x^{\mathcal{C}}_{0opt}$	$y^{\mathcal{C}}_{0opt}$	$z^{\mathcal{C}}_{0opt}$	$d^{\mathcal{C}}_{opt}$
avg	0.052	0.021	1.412	1.414	0.004	0.004	0.031	0.032
std	0.086	0.033	2.208	2.210	0.003	0.001	0.010	0.011

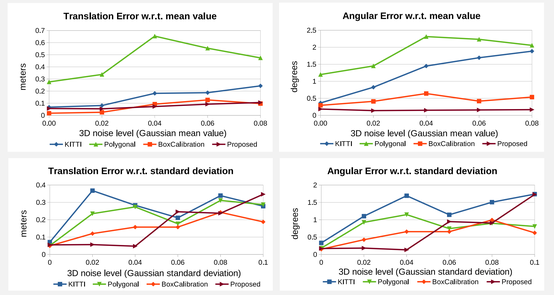
Ellipse fitting test scenes



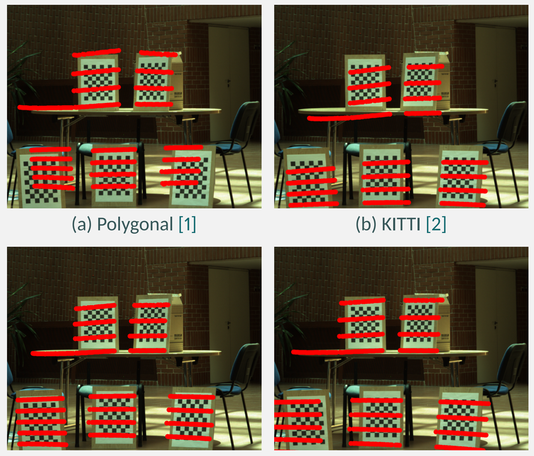
(a) Blender – Simple (b) Blender – Classroom

(c) Carla street-view (d) Real photo

Synthetic tests – *translation and angular error*



Real tests – *reprojection error*



(a) Polygonal [1] (b) KITTI [2]

(c) Box Calibration [3] (d) Proposed

References

[1] Park et al. *Calibration between color camera and 3D LIDAR instruments with a polygonal planar board*, Sensors, 2014

[2] Geiger et al. *Automatic camera and range sensor calibration using a single shot*, IEEE International Conference on Robotics and Automation, 2012

[3] Pusztai et al. *Accurate Calibration of LiDAR-Camera Systems Using Ordinary Boxes*, IEEE International Conference on Computer Vision Workshops, 2017

Contact Information

- E-mail: {tekla.toth, hajder}@inf.elte.hu

Acknowledgements

EFOP-3.6.3-VEKOP-16-2017-00001: Talent Management in Autonomous Vehicle Control Technologies – The Project is supported by the Hungarian Government and co-financed by the European Social Fund.